

MOSAIC-SoC · Block A

One frozen configuration, hardened to a DRC/LVS-clean GF180 macro, and the path that produced it.

1.2488 mm²

1117.5 μ m square macro

DRC · LVS · XOR = 0

full deck, nothing skipped

9/9 corners

timing closed, TNS 0

RE-REVIEW OUTCOME: GO · conditional conditions cleared; closure items open before tapeout

1. From Schematic Review to Layout Review

STATUS

The schematic review flagged one blocker: the deck, config, and target validation described three different chips

What moved since the schematic review

- Physical design 30% → 80%; overall 78% → 87%.
- Block A hardened to a signed-off GF180 macro: DRC, LVS, XOR, antenna all 0.
- RTL frozen and tagged (rtl-freeze-blocka).
- The generator's tapeout gate now describes THIS part; the old PoC no longer qualifies.
- Reproducible on a fresh clone; the flow carries no absolute paths.

The review finding, resolved

- Finding: the deck's 1× FazyRV + 2× SERV die, the 7-hart PoC, and core_registry.TAPEOUT_* were three different chips.
- Now: one config is frozen, declares target: tapeout, and the gate rejects it if any core, memory or platform knob drifts.
- We fixed the config's own header drift: it read "three SERV" over a list of two, the exact thing you flagged.

One question, answered here: which configuration are we taping out, and does it pass physical verification end to end? Every number is a measured signoff metric with the command that produced it.

2. Block A: The One Part Being Taped Out

FROZEN

Neither chip the review saw: a minimal 2-hart SERV SoC that boots and executes from external flash

Aspect	Block A (frozen)
Harts	2 × SERV: TITAN (rv32ic, CSRs, compressed) + worker (rv32i, no CSRs, boot 0x40010000)
Memory	no SRAM pool · 128 B scratchpad (flip-flops) · 1 KB boot ROM
Code	XIP: executes in place from external QSPI flash
Bus / dispatch	OBI crossbar · TDU enabled, dynamic
Peripherals	UART only
Removed	DMA · debug/JTAG · PLIC · SPI host · AO timer · fast-intr · GPIO

Why it looks like this

- Boot hart is SERV, so the FazyRV boot question is moot; it is not in this part.
- The worker boots dormant; the TDU wake is its only release path.
- status_o[6:0] is the only observability with debug: false.
- All 11 other core IPs stay in the generator; none is in this part.

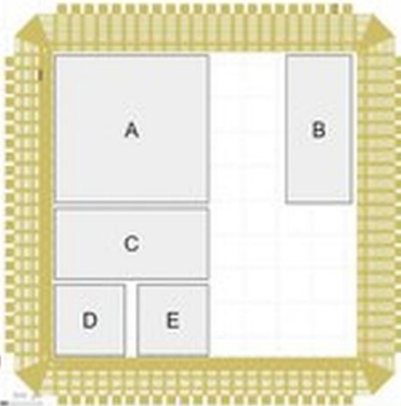
Functional verification on this exact config: boot ROM → QSPI XIP → TDU programming → worker wake → EXIT SUCCESS (both harts report); a separate UART test checks FIFO depth, polled TX, and RX loopback.

3. The Shared MPW Die: Where Block A Sits

MPW

Block A ships as a hard macro into one slot of a shared multi-project die and inherits the project pad ring

- 88 pins
- Total project area 2235um x 2235um
- 5 block configurations
 - A: 1/4 - 22 pins (4 possible locations)
 - B: 1/8 - 16 pins (4 possible locations)
 - C: 1/8 - 6 pins (8 possible locations)
 - D: 1/16 - 10 pins (4 possible locations)
 - E: 1/16 - 6 pins (8 possible locations)
- Separate power for each block
- Chip wide common ground
- Each project specifies i/o pad types
- Placement fixed for each block
- Designs connect to pads when placed
- Smaller projects may be combined into one block. Needs a person to manage block integration



Shared-die plan: 2235 × 2235 μm · 88 pins · 5 block slots (A–E)

Block A's slot

- 1117.5 × 1117.5 μm, one quarter of the 2235 μm die.
- 22-pin budget (20 signal + VDD + VSS).
- Delivered as a hard macro with NO pad ring of its own.
- Separate power per block; blocks share the common ground.
- pad_cfg.py and slot_mosaic.yaml are chip-level artifacts Block A does not use; both now say so in their headers.

4. Fitting the 1.25 mm² Budget

AREA

Area fell from 3.903 to 1.249 mm² (−68%) through measured cuts. The biggest lever: leaving memory off-chip



Why no on-chip SRAM

- GF180's largest SRAM macro holds 512 B and costs 0.209 mm².
- On-chip SRAM costs 0.419 mm² per KB. At 32 KB (the old target) that is 13.4 mm², over 10× the whole budget.
- Code executes XIP from external QSPI flash; on-chip holds a 128 B scratchpad and 1 KB boot ROM.

Where the die goes

- The two SERV cores take ~2% of the die; the platform around them is the area.
- SERV is 8.7× smaller than FazyRV (0.021 vs 0.183 mm²), which is why the boot hart is SERV.
- Biggest cuts: DMA −0.355, debug+PLIC −0.512, full SPI host → XIP reader −0.69 mm².

5. Post-Synthesis Area

SYNTH

Standard-cell area straight out of Yosys, before placement fill and the clock tree. The figure the reviewer asked for

0.855 mm²

post-synthesis cell area

33 852

standard cells

5 587

flip-flops (sequential)

From synthesis to the die

- Yosys synthesis: 854,954 μm^2 of standard cells (33,852 cells, 5,587 flops).
- Place & route sizes and buffers it to 0.987 mm² of std-cell area.
- Hardened die: 1.2488 mm² at 84.4% utilization.

What the area is made of



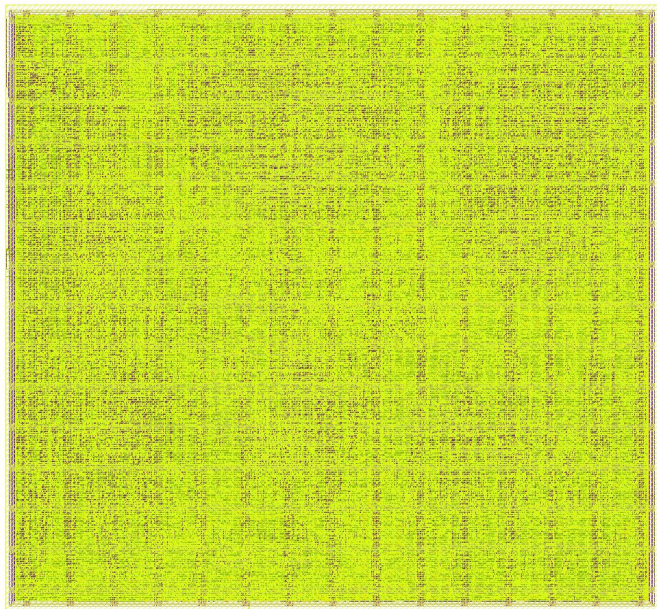
- Flop-heavy by design: the 128 B scratchpad is flip-flops (1,024 bits), plus the two SERV register files and control.
- Flip-flops alone (dffq) are ~0.26 mm², the single largest cell type.

Source: runs/blocka_signoff/06-yosys-synthesis/reports/stat.rpt (GF180MCU 7-track, 5 V library).

6. The Layout: Signed-Off GF180 Macro

LAYOUT

KLayout render of the routed, DRC/LVS-clean Block A macro. The real GDS lives in flow/.../runs/blocka_signoff/final/



Die

$1117.5 \times 1117.5 \mu\text{m} = 1.2488 \text{ mm}^2$

Std cells

44 355 placed · 84.4% utilization

Metal

GF180MCU 5-layer (Metal1–Metal5)

Power

5 V · uniform PDN mesh, 0 grid violations

Clock

10 MHz (100 ns), conservative on purpose

Macros

none bound (no SRAM or analog macro)

7. The 22-Pin Interface: LVS-Verified

PINS

Netgen matched this pin list against the extracted layout and reported "circuits match uniquely", all pins are digital

#	Pin	Dir	Function
1	clk_i	in	clock
2	rst_ni	in	reset, active-low
3	boot_select_i	in	boot source
4	execute_from_flash_i	in	XIP enable
5	uart_rx_i	in	UART receive
6	uart_tx_o	out	UART transmit
7	spi_flash_sck_o	out	QSPI clock
8	spi_flash_cs_o	out	QSPI chip-select

#	Pin	Dir	Function
9–12	spi_flash_sd_io[3:0]	inout	QSPI data (true tristate)
13	status_valid_o	out	exit-value strobe
14–20	status_o[6:0]	out	exit value (only observability)
21	VDD	pwr	5 V
22	VSS	pwr	ground

status_o[6:0] must be bonded. With debug: false and no JTAG, it is the part's only external observability. It drives status_o / status_valid_o from the soc_ctrl exit register.

8. Signoff Results: Nothing Skipped

SIGNOFF

flow/librelane/experimental/run_signoff.sh. The runner refuses to start if its config contains a step substitution

Check	Result	Check	Result
Magic DRC	0	Netgen LVS	match uniquely
KLayout DRC	0	LVS unmatched dev/net/pin	0 / 0 / 0
KLayout↔Magic XOR	0	Antenna	0
Routing DRC	0	Disconnected pins	0
Magic illegal overlap	0	Power-grid violations	0 (VDD, VSS)
Worst IR drop	120 μ V on 5 V	Std-cell utilization	84.4%

LVS ran against the extracted layout, not the wrapper. Netgen reported "Cell pin lists are equivalent" and "Circuits match uniquely": 0 unmatched devices, nets, or pins on the 22-pin contract. None of DRC, LVS, XOR, antenna, or the power grid carries a waiver; all returned zero without one.

9. Timing Closure: All Nine Corners

TIMING

Setup and hold stay positive at every process/voltage/temperature corner, TNS 0. The part is nowhere near its limit

+20.86 ns

worst setup slack

+0.066 ns

worst hold slack

0.0 ns

total negative slack

9 / 9

corners closed

Clock target: 10 MHz (100 ns)

- Below the 50 MHz on the old slide and the padding's 25 MHz.
- 20.7 ns of setup slack leaves large headroom to raise it.
- Frequency can go up once we repair the electrical violations (next slide); a target with the organizers is still open.

The one thin margin

- Hold is positive everywhere but only +66 ps.
- PL_RESIZER_HOLD_SLACK_MARGIN 0.05 traded hold margin for area.
- Re-check before raising the clock; it is listed as an open item.

10. Power

POWER

Vectorless STA estimate at 10 MHz, 5 V. The clock tree and flip-flops dominate; leakage is negligible

58.4 mW

total, nominal corner

73.4 mW

worst of nine corners

7.9 μ W

leakage (negligible)

Where the power goes



- Flip-flops 29.6 mW · clock tree 28.6 mW · combinational 0.25 mW (<1%).
- The flop-heavy scratchpad and the clock net set the power, as they set the area.

How it was measured

- OpenSTA report_power at 10 MHz with default activity, no workload VCD.
- Internal 78% · switching 22% · leakage ~0% (a strength of 180 nm).
- Worst corner is fast / cold / 5.5 V (73.4 mW); best is slow / hot / 4.5 V (47.1 mW).
- Workload-annotated power needs activity capture, which is on the roadmap.

The PDN carries it cleanly: 0 power-grid violations, 120 μ V worst IR drop on 5 V.

11. Waivers: The Two We Accept

WAIVERS

The review asked for no inherited or blanket waivers; DRC, LVS, XOR, antenna and the power grid carry none

Setting	Status	Why it is acceptable
ERROR_ON_SYNTH_CHECKS	ON (was off)	Turning it back on exposed bugs 28–29. No longer waived.
ERROR_ON_MAGIC_DRC	ON	A real violation fails the flow.
ERROR_ON_UNMAPPED_CELLS	ON	An unmapped cell has no physical master.
GRT_ALLOW_CONGESTION	active	Routing then converged to 0 DRC; nothing waived in the result.
max-slew = 591	accepted, bounded	At ss_125C_4v50 only; TT and FF corners clean. The gap to a tapeout-ready macro.
max-fanout = 1	accepted, bounded	Down from 411; one net across all corners; scoped to this design.

Scoping is enforced. Block A's 591-slew / 1-fanout waivers did not carry to Block B: they were "recorded for a different design," so each design reports on its own numbers. Electrical waivers do not travel.

12. What Must Close Before Tapeout

[OPEN](#)

The re-review cleared the conditions and gave a Go; these are the reviewer's items to close before the tapeout itself

MUST CLOSE

- Lock the clock target with the track leads and integration team now, then re-harden at that period. Every STA and slew number depends on it. 25 MHz looks reachable: the worst setup path is a half-cycle output to spi_flash_sd_io, ~3.7 ns of logic.
- Close 591 max-slew + 1 max-fanout. They sit only at ss_125C_4v50; TT and FF corners are clean. Per-net: right-size the violating drivers, try higher-metal / NDR routing; watch fit at 84.4% util.

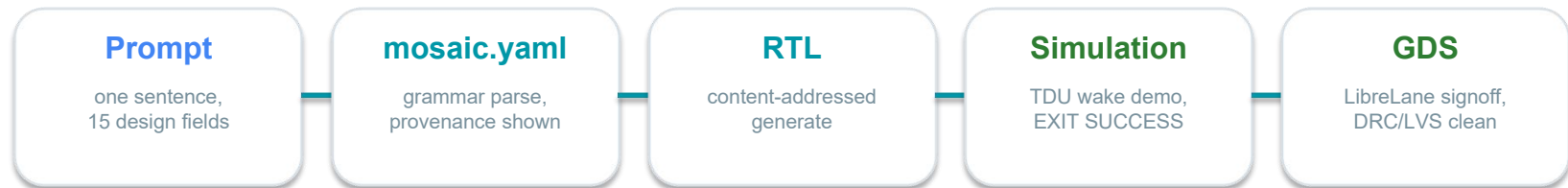
RAISE WITH THE ORGANIZERS

- Timing-annotated GLS: the GF180 cell models use ifnone on edge-sensitive specify paths, which IEEE 1364-2005 §14.2.6 forbids. Confirm whether timing-GLS is a required signoff item or multi-corner STA suffices, and report the models upstream.
- Then tag the re-hardened commit. PHYSICAL_BUNDLE attestation is the project-internal companion to that tag.

13. End-to-End: Prompt to GDS

E2E

Natural-language intent authors the config; deterministic Python generates and checks it. It reached this exact part



Done once, end to end: Block A

- demo/03_blocka_from_prompt.sh: a live model reaches the frozen config through typed CLI arguments, 15 of 15 fields.
- Steps 2–5 call no model (a regex grammar), so they run in CI and gate false 'tapeout' claims.
- Step 6 is the model's work, gated on a harness being installed, not on an API key.

The asymmetry we are honest about

- prompt → RTL → simulation is a GENERATOR: a pairwise covering array, 765 tests, over the whole design space.
- RTL → GDS was, for Block A, a hand-tuned artifact for one design; six experimental configs record how many tries it took.
- Closing that gap is the next slide.

14. Prompt to Signoff

PROMPT

One sentence to a signed-off GDS. The hardening config is derived from the SoC config, with no hand-written physical file

```
# the prompt (demo/03_blocka_from_prompt.sh)
PROMPT="a tapeout SoC: one serv titan
rv32ic +CSR, one serv worker rv32i
boot 0x40010000, no sram, 128B
scratchpad, 1KB rom, XIP flash, uart,
TDU dynamic; no dma/debug/plic/gpio"

# 1 · sentence -> config
./oh-my-soc soc-from-prompt run "$PROMPT"
# -> configs/blocka.yaml (15/15 fields)

# 2 · config -> verified RTL
make mosaic-gen MOSAIC_CFG=configs/blocka.yaml
bash tb/mosaic_soc/run_generic.sh
# -> EXIT SUCCESS

# 3 · config -> signed-off GDS (derived)
MOSAIC_HARDEN_FROM_SOC=configs/blocka.yaml \
flow/librelane/experimental/run_signoff.sh
# -> DRC / LVS / XOR / antenna = 0
```

What this proves

- Step 1 is a deterministic grammar. demo/03 also shows a live model reach the frozen config, 15 of 15 fields.
- Step 3 derives the floorplan and PDN from the config. No hand-written physical file is in the chain.
- Proven: Block B hardened from a generated config, clean; Block A's config regenerates key-for-key.

Correctness (DRC, LVS, timing) is generated end to end. Electrical polish, the max-slew work, is still per-design.

15. Derive, Don't Fit: Generating the Hardening

FRONTIER

The hand-fitting is only the floorplan; PDN and timing setup are design-independent, so we predict the die, then generate it

	Block A	Block B	Note
Die side (μm)	1117.5	1261.6	predicted 1223 $\rightarrow$ +3.2% error
Die area (mm^2)	1.2488	1.5917	+27.5% for +1 worker
Max-slew viol.	591	845	grows faster than area (+43%)
Hardening config	hand-tuned	GENERATED	clean on every hard check

What is now proven

- Die size is DERIVABLE: the area model lands within 0.1–0.4% of measured.
- Block B hardened from a GENERATED config (no hand-written physical file): DRC/LVS/XOR/antenna 0, setup +20.89 ns.
- Block C: prediction registered BEFORE the run.

What it revealed

- Electrical violations grow FASTER than area, so they are not a fix-once cost.
- Waiver scoping held: Block A's waivers stayed off Block B.
- +5.3 pts utilization bought –7.6% area but +73% slew: a quality trade, now made in the open (default utilization 0.80 $\rightarrow$ 0.75).

16. Verification & Signoff Checklist

CHECK

Required physical-review categories mapped to Block A evidence. PASS means a reproducible artifact, not a plan

Physical DRC

Magic 0 · KLayout 0 · routing 0 · XOR 0

PASS

LVS

Netgen "match uniquely"; 0 errors · 0 property fails

PASS

Antenna

0 violations

PASS

Power / IR

0 grid violations · 120 μ V IR · ~58 mW

PASS

Timing (STA)

9/9 corners, setup +20.86 / hold +0.066

PASS

Functional GLS

post-route, EXIT SUCCESS, 12,399 cycles

PASS

Electrical (slew)

591 at ss_125C_4v50 only; TT/FF clean

PARTIAL

re-harden

re-harden at locked clock

PARTIAL

Timing-annot. GLS

GF180 ifnone models; ask organizers

OPEN

Tapeout attestation

PHYSICAL_BUNDLE; tag re-hardened commit

OPEN

17. Re-Review Outcome: GO

REVIEW

The reviewer returned Go: conditional conditions cleared and the committed signoff run verified, with items open before tapeout

GO

The reviewer verified the committed signoff run; the main checks pass. Specific closure items remain before tapeout.

What the reviewer confirmed

- Config consistency resolved; the 22-pin interface matches LVS.
- Post-route functional GLS: EXIT SUCCESS, 12,399 cycles, through the bonded interface only.
- DRC, LVS, XOR, antenna 0; timing clean at 9 corners; no waivers on DRC / LVS / XOR / antenna / PG.
- Two functional bugs the flow exposed (soc_ctrl DMA tie-off, SERV Wishbone tie-off) fixed and regression-covered.
- "Clean DRC / LVS at 84.4% util is solid P&R, not a trade-off."

Gate to tapeout — must close

- Lock the clock with the track leads + integration now, then re-harden at that period. 25 MHz looks reachable (worst logic ~3.7 ns).
- Close 591 max-slew + 1 fanout — only at ss_125C_4v50; TT and FF corners clean. Per-net driver sizing, not blanket buffers.
- Settle the timing-annotated GLS question with the organizers (GF180 ifnone models vs IEEE 1364).
- Then tag the re-hardened commit.

Appendix · Reproduce the Signoff

[REPRO](#)

Three commands on a fresh clone, and the exact toolchain. LibreLane 3.0.0 Nix-pins every tool below it

Step by step (from a clean checkout)

```
# 1 · generate the RTL bundle
make mosaic-gen \
  MOSAIC_CFG=configs/\
  mosaic_tapeout_ultra.yaml

# 2 · functional check (both harts)
MOSAIC_CFG=configs/mosaic_tapeout_\
ultra.yaml \
  bash tb/mosaic_soc/run_generic.sh
# -> EXIT SUCCESS

# 3 · harden + full signoff (hours)
cd flow/libreLane && \
  ./experimental/run_signoff.sh
# -> DRC/LVS/XOR/antenna = 0
```

The checked-in LibreLane configs carry no absolute paths; gen_filelist.py resolves the source list from the FuseSoC manifest at run time.

Tools & PDK (pinned)

Component	Version
LibreLane (flow)	3.0.0 · Nix-pinned
Yosys (synthesis)	0.62
OpenROAD (P&R, STA)	pinned by LibreLane 3.0.0
Magic (DRC, extract)	8.3
KLayout (DRC, XOR)	pinned by LibreLane 3.0.0
Netgen (LVS)	1.5
Verilator (sim)	5.044
PDK	gf180mcuD
Std cells	gf180mcu_fd_sc_mcu7t5v0 (7T, 5 V)

Reproduces the re-hardened result, a byte copy of the exact GDS on slide 6.